Shubham **Kejriwal**

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Professional Summary

Final year Ph.D. candidate studying gravitational wave sources for the upcoming LISA mission. Experience in the LISA data analysis pipeline, modeling and inference of astrophysical environments and modified-gravity effects, waveform development, and search for electromagnetic counterparts of extreme-mass-ratio inspirals. Proficient in Bayesian statistics and machine learning with a working knowledge of general-relativistic waveform modeling.

Education

National University of Singapore (NUS) — *Singapore*
Ph.D. Physics — *2023 — Present*

- Gravitational Waves Group, Department of Physics, Faculty of Science, NUS
- Supervisor: Prof. Alvin J. K. Chua

Shiv Nadar University — *Delhi, India*
B.Sc. (Research) in Physics with a Minor in Mathematics (distinction) — *2018 — 2022*

- Undergraduate Thesis: "On the Effects of Relative Motion on Gravitational Waves from Extreme-Mass-Ratio Inspirals".
- Supervisor: Prof. Alvin J. K. Chua

Publications

Total *8* articles available as preprints or publications. Also see inspire-HEP and arXiv.

- **S. Kejriwal**, E. Barausse, A. J. K. Chua (2025). *Hierarchical modeling of gravitational-wave populations for disentangling environmental and modified-gravity effects*; (arXIv)
- J. Chakraborty, L. V. Drummond, M. Bonetti, A. Franchini, **S. Kejriwal**, et al. (2025). *Prospects for EMRI/MBH parameter estimation using Quasi-Periodic Eruption timings*; (ApJ)
- C. E. A. Chapman-Bird, L. Speri, Z. Nasipak, O. Burke, M. L. Katz, A. Santini, **S. Kejriwal**, et al. (2025). *Efficient Waveforms for asymmetric-mass eccentric equatorial inspirals into rapidly-spinning black holes*; (PRD)
- **S. Kejriwal**, F. Duque, A. J. K. Chua, J. Gair (2025). *Bias-Corrected Importance Sampling for Inferring Beyond Vacuum-GR Effects in Gravitational Wave Sources*; (PRD)
- F. Duque, **S. Kejriwal**, L. Sberna, L. Speri, J. Gair (2024). *Constraining accretion physics with gravitational waves from eccentric extreme-mass-ratio inspirals*; (PRD)
- D. R. Pasham, **S. Kejriwal**, E. Coughlin, V. Witzany, A. J. K. Chua, M. Zajaček, T. Wevers, Y. Ajay (2024). *Alive and Strongly Kicking: Stable X-ray Quasi-Periodic Eruptions from eRO-QPE2 over 3.5 Years*; (arXiv)
- **S. Kejriwal,** V. Witzany, M. Zajaček, D. R. Pasham, A. J. K. Chua. (2024). *Repeating Nuclear Transients as Candidate Electromagnetic Counterparts of LISA Extreme Mass Ratio Inspirals*; (MNRAS)
- **S. Kejriwal,** L. Speri, A. J. K. Chua. (2024). *Impact of Correlations on the Modeling and Inference of Beyond Vacuum-GR Effects in Extreme-Mass-Ratio Inspirals*; (PRD)

Talks

Invited

- Institute of Physics Singapore (IPS) Meeting, NUS, Singapore. September 2025.
- Astroparticle Physics Group Seminar Talk, SISSA, Trieste, Italy. September 2025.
- Gravitational Physics Group Talk, ETH Zurich, Switzerland. September 2025.
- Astroparticle and Cosmology Laboratory (APC) Seminar Talk, Paris, France. July 2025.

- 245th American Astronomical Society Meeting, Washington D.C., Special Session on Repeating Transients, (invited review talk, declined). January 2025.
- LISA Meeting Talk, Max Planck Institute for Gravitational Physics (AEI), Potsdam, Germany. July 2024.
- 17th Marcel Grossmann Meeting (MGXVII), The 'Gabriele d'Annunzio' University, Italy (declined). July 2024.
- LIGO Seminar Talk, Caltech, California, USA. April 2024.
- Gravity Theory Group Seminar Talk, University of Illinois, Urbana-Champaign (UIUC), Illinois, USA (remote). January 2024.

Contributed

- EMRI Search and Inference within the LISA Global Fit - Part I, University of Paris, Paris, France. June 2025.
- 15th International LISA Symposium 2024, University College, Dublin, Ireland. July 2024.
- 27th Capra Meeting on Radiation Reaction in General Relativity, NUS, Singapore. June 2024.
- Amaldi15 - Premiere International Conference on Gravitational Waves (remote). July 2023.
- 1st Trieste Meeting on the Physics of Gravitational Waves, SISSA, Trieste, Italy. June 2023.

Awards

- NUS Department of Physics Best Graduate Researcher Award (2025).
- NUS Research Scholarship (2023-2026).
- Dean's List Academic Excellence Award (x3), Shiv Nadar University (2019, 2020).

Teaching and Mentoring

Teaching and Teaching Assistance

- HS1502 "Conceptual Introduction to Machine Learning", NUS.
  (2023-24: Semester 2, 2024-25: Semester 1).
- COS1000 "Computational Thinking for Scientists", NUS.
  (2023-24: Semester 1 and 2, 2024-25: Semester 1).
- PC5252 "Bayesian Statistics and Machine Learning", NUS.
  (2023-24: Semester 1, 2024-25: Semester 2, 2025-26: Semester 2).

Student Mentoring

- Ms. Swathi Ganesh, B.Sc. Physics, Shiv Nadar University (2025).
- Ms. Sheena Abigail, M.Sc. Physics, NUS. Final year project (2024-25).

Contributions

Posters

- Faculty of Science 95th Anniversary Symposium, Department of Physics, NUS, Singapore. September 2024.
- Physics Engagement Camp, Department of Physics, NUS, Singapore. May 2024.

Discussions

- Enabling Future GW Astrophysics in the mHz Regime, MIAPbP, Garching, Germany. "Including environmental effects in GW waveforms" (co-chair). July 2025.
- EMRI Search and Inference within the LISA Global Fit - Part I, Astroparticle and Cosmology Laboratroy (APC), Paris, France. "Data Analysis Strategies" (co-chair). June 2025.
- 27th Capra Meeting on Radiation Reaction in General Relativity, NUS, Singapore. "Priorities for EMRI science" (co-chair). June 2024.

Collaborations

- LISA Consortium (core member) — Waveform WG, Data Analysis WG, Fundamental Physics WG; LISA Asia Member.
- LISA Distributed Data Processing Centre (full member).
- `StableEMRIFisher(SEF)` (core developer).
- `FastEMRIWaveforms(FEW)` (core developer).

Participation

- Mathematical Methods for the General Relativistic Two-body Problem, August 2025, NUS, Singapore.
- Enabling Future Gravitational Wave Astrophysics in the Milli-Hertz Regime, July 2025, MIAPbP, Munich, Germany.
- EMRI Search and Inference within the LISA Global Fit - Part I, June 2025, Astrophysics and Cosmology Laboratory (APC), Paris, France.
- Beyond the Horizon: Testing the Black Hole Paradigm, March 2025, ICTS-TIFR, Bengaluru, India.
- LISA Sprint Workshop, April 2024, Caltech, California, USA.

Others

- Referee for Physical Review D.
- EMRI Search and Inference within the LISA Global Fit - Part II (2026), Local Organizing Committee.
- 27th Capra Meeting on Radiation Reaction in General Relativity (2024), Local Organizing Committee.
- Annual Physics Conference (2023 and 2025), Department of Physics, NUS, Singapore, Organizing Committee.